

Aashish Sheshadri

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PROFILE

Twelve years building the layer that lets automated judgment reach production safely — currently the agentic SDLC platform; before that the company-wide standard governing how much autonomy an ML system is granted, and hierarchical reinforcement learning for cluster capacity. Individual contributor: sets technical direction across orgs with no reporting line. 5 peer-reviewed publications (362 citations, Google Scholar, Aug 2026), 1 issued patent and 1 pending application as lead inventor, and the AAAI HCOMP Test of Time Award for SQUARE — a benchmark for aggregating disagreeing human judgments into reliable evaluation signal, which is the problem that reappears today as annotation quality and model-graded evaluation.

EXPERIENCE

PayPal Inc, San Jose, CA

June 2014 – Present

Machine Learning Architect (Sr. MTS)

Feb 2023 – Present

Set ML and infrastructure strategy across cloud platform, data, and developer productivity. Decide which problems get an ML solution vs. a heuristic, select modeling approaches, and govern production rollouts across ~1,550 applications.

Agentic SDLC — Infra, Application, and Data

- Built the multi-agent platform that carries an engineer's stated intent to production across three lifecycles: **Infra SDLC** (provisioning and environment management), **Application SDLC** (onboarding through per-change environments, functional-test gates, and canary promotion), and **Data SDLC**. Specialist agents behind a single orchestrator, sharing memory, policy, and a skill catalog. In production across all three lifecycles since August 2026; adoption and cycle-time numbers still being collected.
- Set the safety envelope, which is where the design effort went: no agent holds a write path, and every mutation is executed by deterministic code acting on an agent's proposal. Authority is budgeted by blast radius — an open-ended tool loop for the agent furthest from production, a short schema-validated pipeline for the one that touches it.
- Made promotion a code check rather than an instruction: an evidence-first evaluator scores acceptance criteria and a state transition keyed on that verdict holds the change in draft until it passes.

ML systems and deployment governance

- Designed the ML deployment framework — canary evaluation, shadow mode, **graduated autonomy** — now the company-wide production standard governing how much autonomy an ML system is granted. Introduced as a Staff engineer.
- Defined what evidence a system must produce before its autonomy widens, so the question is answered by a gate rather than a review meeting.
- Set autoscaling and capacity strategy, and built the hierarchical RL policy for cluster capacity allocation across ~1,550 applications (~100k pods), taking the demand forecasts from the prior role as its input. The hard part was reward function and curriculum engineering, not the policy architecture.
- Found and closed reward-specification failures under optimization pressure — cases where the policy optimized the written objective into an outcome I had not intended. Reward design, and the evaluation that catches where a reward diverges from its intent, is the part of this work that transfers.
- Ongoing collaboration with IBM Quantum on variational algorithms for risk modeling (published in *Quantum*, 2023).

Staff Machine Learning Engineer (MTS 2)

Mar 2020 – Feb 2023

- Led ML-driven predictive autoscaling for ~1,500 applications, working toward five-nines availability while cutting compute spend ~25%.
- Carried the sequence-forecasting work from infrastructure signals to demand: time-series models with confidence-calibrated predictions for ~1,500 applications, handling distribution shift across seasonal, promotional, and organic traffic patterns. Those forecasts became the signal the autoscaler acted on rather than a dashboard nobody read.
- Developed sequence classification models using BERT for real-time threat detection in information security — security-critical inference where false negatives are expensive.
- Collaborated with IBM Quantum on variational quantum algorithms for feature selection in credit risk modeling. Published in Quantum (2023).

Senior Machine Learning Engineer (MTS 1)

Mar 2018 – Feb 2020

- Applied sequence-to-sequence models (RNN/LSTM) to CPU and memory forecasting across ~350,000 monitored signals per second, replacing static alert thresholds with likelihood-based outlier detection. Presented at QCon.ai 2019 — infoq.com/presentations/journey-99999
- Architected the enterprise ML platform: multi-tenant GPU scheduling with prioritized access across business units, remote kernel orchestration supporting PyTorch, H2O, R, and Ray workloads, and isolated environments from notebook experimentation through production model serving.
- Built pipeline orchestration layer: notebook scheduling via Airflow, automated training pipelines with managed retraining cadences, and DAG-based workflow composition.
- Built the parameterized-notebook core behind PPExtensions' `run_pipeline` magic (PayPal open-source, BSD-3 — github.com/paypal/PPExtensions): the command parser and the AST-level parameter substitution that type-checks and injects values into each notebook's parameter cell, which is what lets a pipeline hand state from one stage to the next. Also the CSV connector, logging and configuration layers.

Senior Research Engineer

Mar 2016 – Feb 2018

- Co-authored the FASH64 and FASH256 specifications with Doug Crockford (JSON inventor) — fast non-cryptographic hash functions validated by statistical dispersion analysis and empirical collision-rate testing (crockford.com/fash64.pdf).
- Designed and open-sourced the **Seif Protocol** — an ECC-512 handshake over Node.js, MIT-licensed (github.com/paypal/seif-protocol) — protocol design, formal verification, and cryptographic systems engineering from first principles.
- Research in probabilistic analysis and randomness validation — mathematical foundations later applied to distribution modeling and uncertainty quantification in ML systems.

Research Engineer

Jun 2014 – Feb 2016

- Built and open-sourced the cryptographic foundation of the Seif project under Doug Crockford, both MIT-licensed: **seifnode**, the Node.js native addon exposing ISAAC RNG, ECC-512, AES-GCM and SHA3-256 (**543 stars** — github.com/paypal/seifnode), and **seifrng**, the C++ entropy-mining and CSPRNG layer beneath it (github.com/paypal/seifrng). Presented at OSCON in 2015 and 2016.

The University of Texas at Austin, Austin, TX

Sept 2012 – May 2014

Graduate Research Assistant

- Research in IR evaluation and human computation: aggregating noisy, disagreeing human judgments into reliable evaluation signal, and hybrid expert/non-expert judging for building test collections affordably — the evaluation-quality problems that recur today in annotation aggregation and model-graded evaluation.
- Built and open-sourced the SQUARE benchmark (MIT) — a common harness for comparing crowd-consensus algorithms (majority vote, GLAD, CUBAM, Dawid-Skene) under supervised, semi-supervised, and unsupervised

regimes: github.com/utir/square, and a second-generation Java implementation adding n-fold evaluation and a worker-simulation framework at github.com/utir/square-2.0.

Teaching Assistant

Sept 2012 – Dec 2012

- Programming Languages (undergraduate course)

Carnegie Mellon University, Pittsburgh, PA

Sept 2011 – May 2012

Research Associate

- LiDAR-camera calibration for mapping and localization. Visual odometry and orthographic views for global localization of planetary rovers. Contributed to Astrobotic's Google Lunar XPRIZE effort.

EDUCATION

The University of Texas at Austin — M.S. Computer Science, GPA 3.90

Aug 2012 – May 2014

Computer Vision & IR Thesis Track

Carnegie Mellon University — Audited Courses in Computer Vision, Robotics & Perception

Aug 2011 – May 2012

PES Institute of Technology, Bangalore — B.S. Electronics & Communication, GPA 8.40, Honors

Jun 2007 – Aug 2011

PUBLICATIONS

- Variational quantum algorithm for unconstrained black box binary optimization: Application to feature selection — *Quantum*, Jan 2023
- Mix and match: collaborative expert-crowd judging for building test collections accurately and affordably — *DESIRES*, Aug 2018
- A collaborative approach to IR evaluation (Masters Thesis) — *UT Digital Repository*, May 2014
- SQUARE: A Benchmark for Research on Computing Crowd Consensus (First Author, **Test of Time Award, Oct 2024**) — *AAAI HCOMP*, Nov 2013. Code: github.com/utir/square
- Position Estimation by Registration to Planetary Terrain (First Author) — *IEEE MFI*, Sept 2012
- Complementary Flyover and Rover Sensing for Superior Modeling of Planetary Features — *FSR*, July 2012

PATENTS

- **Session Management System** (Lead Inventor) — US11818219, Issued Nov 14, 2023
- **Dynamic Autoscaling of Server Resources Using Intelligent Demand Analytic Systems** (Lead Inventor) — application US 17/826,071, filed May 26, 2022 (pending)

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript; Go, C, SQL

Frameworks: PyTorch, TensorFlow, Kubernetes, GCP/AWS/Azure, Spark, Ray, Jupyter

Domains: Agentic Systems & Tool-Use Harnesses, Reward & Curriculum Design, Evaluation Methodology, Reinforcement Learning, Deployment Safety & Oversight, ML Infrastructure, Time-Series Forecasting

INVITED SPEAKER

QCon, Strata, MLConf, Deep Learning World, RE•WORK Enterprise AI Summit, OSCON, Fluent

Recorded: *On a Deep Journey towards Five Nines* — QCon.ai 2019, infoq.com/presentations/journey-99999

PEER REVIEWER

Information Processing & Management (IF 7.4), Artificial Intelligence (IF 5.1)